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THIRD SEMESTER B.Tech EXAMINATION DECEMBER – 2019

Name of the Student :

SET- B

III SEMESTER MCQ – I

Subject: Diploma Mathematics-I		Session :	Jul-Dec 2019
Subject Code: 18MADIP01		Date of Examination :	
Signature of Faculty Incharge with date		Marks Obtained	/40

1. Particular integral of the differential equation $(D^2 + D - 2)y = e^x$

a) $\frac{xe^x}{3}$ b) $\frac{xe^2}{4}$ c) $\frac{xe^2}{3}$ d) $\frac{x^2e^2}{3}$

2. $\sinh ax = ...$

a) $\frac{e^{ax} + e^{-ax}}{2}$ b) $\frac{e^{ax} - e^{-ax}}{2}$ c) $\frac{e^{ax} + e^{-ax}}{3}$ d) $\frac{e^{ax} + e^{-ax}}{4}$

3. Roots of the auxiliary equation of differential equation $(D^2 - 6D + 9)x = 5e^{-2t}$ are ....

a) 3,–3 b) 2, 3 c) –3, 2 d) 3, 3

4. A pair of linear equations to determine value of two variables is called

a) Simultaneous linear equations b) Paired equations c) Quadratic equations d) Simple equations

5. If $\frac{d^2y}{dx^2} + y = \tan x$, then $W = ...$ using variation of parameters method

a) 4 b) 2 c) 3 d) 1

6. Complete solution of the differential equation of second order using variation of Parameter method is given by.....

a) $y = Ay_1 + By_2$ b) $y = Ay_1 - By_2$ c) $y = A'y_1 + B'y$ d) $y = A'y_1 - B'y$

7. Substituting $ax + b = e^t$ in Legendre's differential equation, $(ax+b)^2 y'' = ...$

a) $a^3(D-1)(D-2)y$ b) $a^3D(D-1)y$ c) $a^2D(D-2)y$ d) $a^2D(D-1)y$

8. Roots of the auxiliary equation $m^2 - 2m + 1 = 0$ are....

a) 1,-1 b) 1, 1 c) 0, 0 d) 0, 1

9. Nature of the roots $p \pm iq$ is

a) Equal b) Real and Distinct b) Complex d) None of these

10. Cauchy's linear equation is a particular case of Legendre's linear equation when $a = ...$ and $b = ...$

a) 2 and 3 b) 1 and 3 c) 1 and 2 d) 1 and 0

11. $\int e^x \, dx$ is equal to

a) $\frac{e^x}{x} + c$ b) $e^{-x} + c$ c) $e^x + c$ d) None of these

12. $\int \sin x \, dx$ is equal to

a) $\sin x + c$ b) $-\cos x + c$ c) $\cos x + c$ d) $-\sin x + c$

13. If $f(-x) = f(x)$, then $\int_{-a}^{a} f(x)dx$ is equal to

a) $2\int_{a}^{0} f(x)dx$ b) $2\int_{0}^{a} f(x)dx$ c) $-2\int_{a}^{0} f(x)dx$ d) $\int_{0}^{a} f(x)dx$

14. $\int x^n \, dx =$_____

a) $\frac{x^{n+1}}{n+1} + c$ b) $\frac{x^{n-1}}{n-1} + c$ c) $n \, x^{n-1} + c$ d) None of these

15. $\int \frac{x}{x^2+4} \, dx$ is equal to

a) $\log(x^2+4) + c$ b) $\log x + c$ c) $\frac{1}{2}\log(x^2+4) + c$ d) None of these

16. The value of definite integral $\int_{0}^{\frac{\pi}{2}} \sin^4 x \, dx$ is

a) $\frac{3\pi}{16}$ b) $\frac{4\pi}{8}$ c) $\frac{6\pi}{10}$ d) $\frac{7}{10}$

17. The value of definite integral $\int_{0}^{\frac{\pi}{2}} \cos^7 x \, dx$ is

a) $\frac{3\pi}{8}$ b) $\frac{16\pi}{35}$ c) $\frac{2}{7}$ d) $\frac{16}{35}$

18. $\Gamma(n+1)$ is equal to

a) $\Gamma(n)$ b) $n\Gamma(n)$ c) $n\Gamma(n-1)$ d) $\Gamma(n+1)$

19. The valve of $\Gamma(\frac{5}{2})$ is

a) $\frac{4}{5}\sqrt{\pi}$ b) $\frac{5}{6}\sqrt{\pi}$ c) $\frac{3}{4}\sqrt{\pi}$ d) None of these

20. $\beta(m,n)$ is

a) $\frac{\Gamma(m).\Gamma(n)}{\Gamma(m-n)}$ b) $\frac{\Gamma(m)+\Gamma(n)}{\Gamma(m+n)}$ c) $\frac{\Gamma(m).\Gamma(n)}{\Gamma(m+n)}$ d) $\frac{\Gamma(m).\Gamma(n)}{\Gamma(m)}$

21. Gamma Function is defined as

a) $\int_0^\infty e^{-x}x^{n-1}\ dx$, (b) $\int_0^\infty e^{-3x}x^{n-2}\ dx$, (r c) $\int_0^\infty e^{-xy}x^{n}\ dx$, (n d) None of these

22. $\beta(\frac{7}{2},-\frac{1}{2}) =$

a) $-\frac{13}{6}\pi$ b) $-\frac{14}{9}\pi$ c) $-\frac{15}{8}\pi$ d) $-\frac{16}{7}\pi$

23. $\int_0^{\frac{\pi}{2}} \sin^{2m-1}\theta \cos^{2n-1}\theta\, d\theta =$

a) $\beta(m)$ b) $\beta(m+n)$ c) $\beta(n)$ d) $\beta(m,n)$

24. Order and degree of $\left(\frac{d^3y}{dx^2}\right)^4 - 6x^2\left(\frac{dy}{dx}\right)^8 + e^y = \sin xy$ is

a) Order=3 Degree=4 b) Order=4 Degree=4 c) Order=8 Degree=4 d) Order=2 Degree=6

25. Order and degree of $\frac{dy}{dx} + P(x)y = y^nQ(x)$

a) Order=1 Degree=2 b) Order=1 Degree=3 c) Order=0 Degree=0 d) Order=1 Degree=1

26. Solution of $(x^2+1)\frac{dy}{dx}+(y^2+1)=0$ is....

a) $\tan^{-1}x+\sin^{-1}x=c$ b) $\tan^{-1}x+\tan^{-1}y=c$ c) $\tan^{-1}x+\sec x=c$ d) $\tan x+\tan y=c$

27. The standard form of exact differential equation is.....

a) $Mdy+Ndx=0$ b) $Mdy+Ndx=0$ c) $Mdx+Ndy=0$ d) $Mdx+Ndy=0$

28. The necessary and sufficient condition that the differential equation to be exact is

a) $\frac{\partial M}{\partial y}=\frac{\partial N}{\partial x}$ b) $\frac{\partial N}{\partial y}=\frac{\partial N}{\partial x}$ c) $\frac{\partial M}{\partial y}=\frac{\partial M}{\partial x}$ d) $\frac{\partial M}{\partial x}=\frac{\partial N}{\partial y}$

29. $\frac{\partial M}{\partial y}$ & $\frac{\partial N}{\partial x}$ for the differential equation $ydx+(x+y)dy=0$ is

a) 5 b) 1 c) 0 d) None of these

30. Integrating factor of $\frac{dy}{dx}+Py=Q$ is

a) $e^{\int Qdx}$ b) $e^{\int Pdx}$ c) $e^{\int Qdy}$ d) $e^{\int Pdy}$

31. In Bernoulli's equation $\frac{dy}{dx}+Py=Qy^n$ P & Q are functions of _____.

a) P & Q are functions of y b) P & Q are functions of x & y c) P & Q are functions of x d) Both a & b

32. Solution of $ydx=xdy$ is

a) $x^2=cy$ b) $x^2=cy^2$ c) 0 d) $x=cy$

33. Solution of $\sec^2 x\tan ydx+\sec^2 y\tan xdy=0$ is....

a) $\tan x\tan y=c$ b) $\tan x=c$ c) $\tan y=0$ d) $\tan x+\tan y=c$

34. A differential equation is considered to be ordinary if it has.....

a) one dependent variable b) more than one dependent variable c) one independent variable d) more than one independent variable

35. Differential equation is said to be linear if the dependent variable and its derivatives are of

a) First degree b) Second degree c) Third degree d) None of these

36. The general solution is of the form $y=y_c+y_p$, where y_p is called

a) Auxiliary equation b) Particular integral c) Complimentary function d) None of these

37. Suppose the roots of the auxiliary equation are real and distinct(m_1 and m_2)then the complimentary function is of the form

a) $y=c_1e^{m_1x}+c_2e^{m_2x}$ b) $y=c_1e^{m_1x}-c_2e^{m_2x}$ c) $y=c_2e^{m_2x}$ d) $y=c_1e^{m_1x}$

38. If $(D^2-5D+6)y=0$, then the solution of the differential equation is

a) $y=c_1e^{2x}+c_2e^{3x}$ b) $y=c_1e^{2x}+c_2e^{-2x}$ c) $y=c_1e^{-3x}+c_2e$ d) $y=c_1e^{2x}-c_2e^{3x}$

39. $\frac{1}{D}(\cos x)$ is

a) $-\sin x$ b) $\tan x$ c) $\sin x$ d) None of these

40. $\frac{1}{D^2}\left(e^{3x}\right)$ is

a) $\frac{e^{-3x}}{9}$ b) $\frac{e^{3x}}{3}$ c) $\frac{e^{-3x}}{3}$ d) $\frac{e^{3x}}{9}$